

Can Breast MRI Help Predict Treatment Response?

Forecasting pretraining made the vessel map matter, but timing complicates the story

Spencer Venancio

with Sarit Bose and Aakrithi Ram · Advisor: Anna Woodard, PhD

Before surgery, chemotherapy is a high-stakes test

THE TREATMENT

Before surgery

Chemotherapy can shrink a tumor and reveal how it responds.

THE OUTCOME

≈ 1 in 3

scans came from patients with no remaining invasive cancer at surgery.

THE QUESTION

Can MRI help?

Could an existing scan give us an earlier clue about treatment response?

The long-term goal is better evidence for tailoring treatment, not replacing clinical judgment.

Tumors build a messier blood supply

New vessels don't look or behave like normal ones.

NORMAL VESSELS

**Organized,
efficient**

Blood flows through a predictable, well-structured network.

TUMOR VESSELS

**Dense,
tortuous,
leaky**

Structure may reflect how aggressive a tumor is.

OUR IDEA

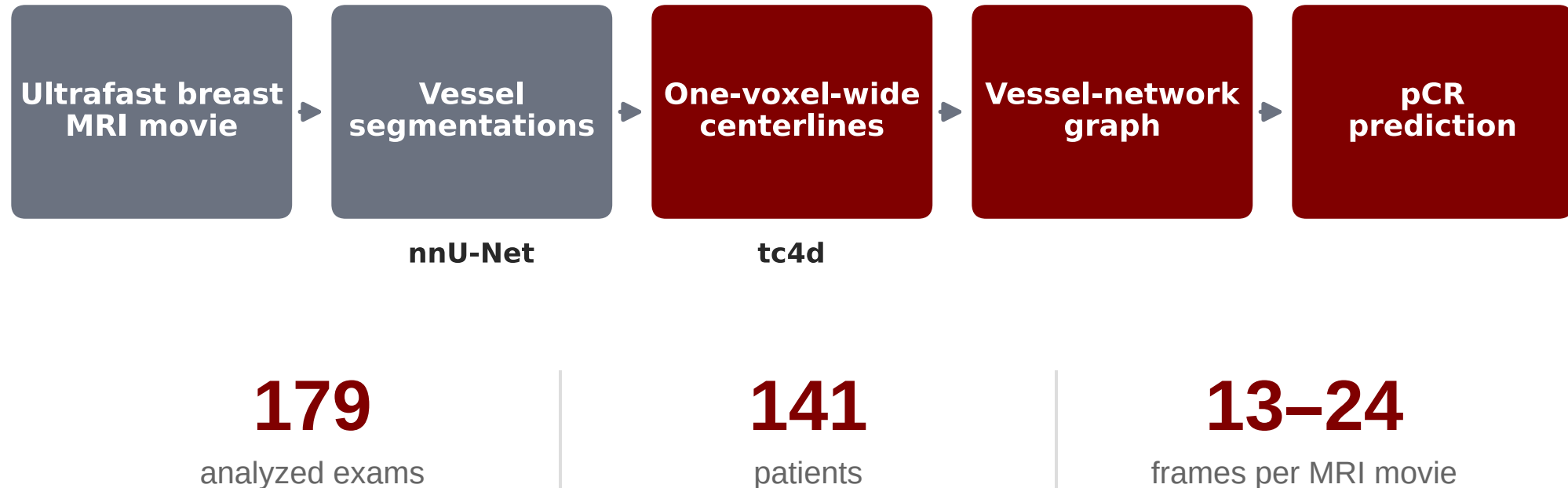
**Read the
vessels**

If that structure tracks tumor biology, it might help predict response.

That's the biological reason to expect a vessel-based signal, not just a modeling choice.

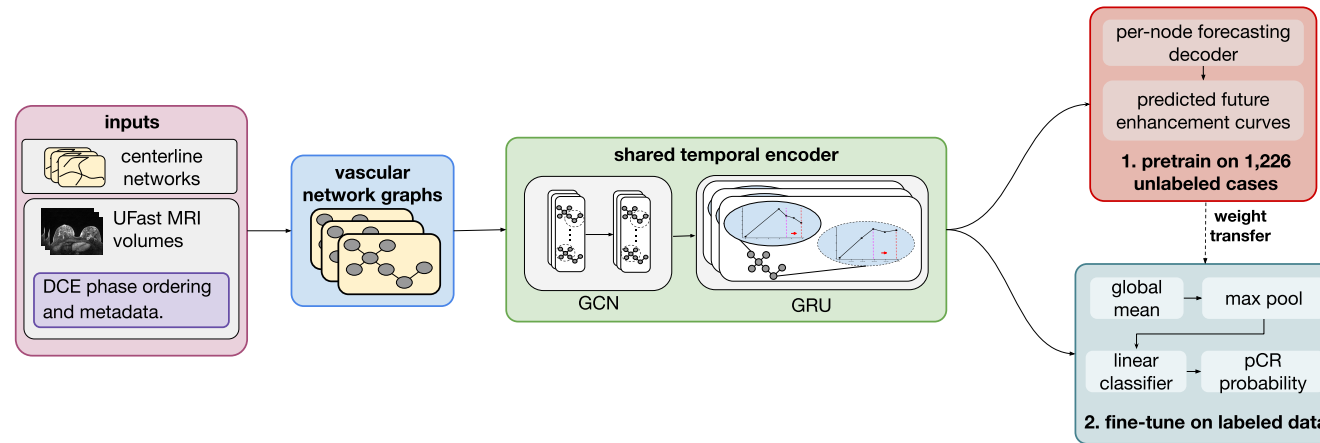
An MRI “movie” shows more than anatomy

After contrast is injected, repeated MRI frames reveal **how it appears through the breast’s vessels over time.**



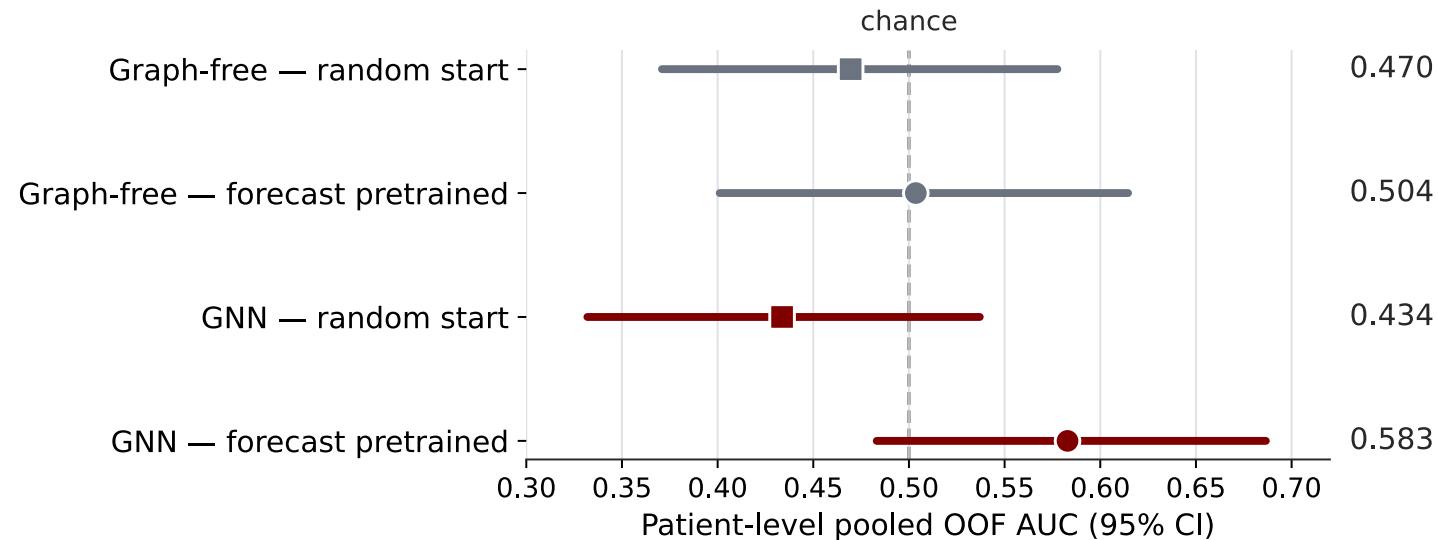
Two ingredients, one controlled comparison

Does **connecting the vessel map** help? Does **rehearsing the future** help?



Encoder pretrained on 1,226 unlabeled exams, then fine-tuned with and without connections, from pretrained or random weights.

Pretraining gave the graph model a real boost



The boost was real: **+0.149 [+0.046, +0.255]** AUC for the GNN, a gain graph-free didn't reliably show. Pretrained, GNN then beat graph-free by **+0.079 [+0.010, +0.155]**.

But the clock may be part of the story

Timing alone scores nearly as well.

A model using only frame count, scan duration, and cadence (no imaging at all) reaches 0.587 AUC, almost identical to the pretrained graph model's 0.583.

The two track each other.

Their predictions correlate (Pearson $r = 0.50$), so acquisition timing is associated with outcome in this cohort.

This raises concern about protocol dependence, but doesn't tell us how much of the signal is technical versus biological. Adjustment could also remove real patient differences.

What should you remember?

1

Forecast pretraining gave the **graph model a real boost**, which then beat the matched simple-summary model.

2

Vessel enhancement still carries only **modest, uncertain** signal on its own.

3

Acquisition timing tracks outcome too. Next, we **separate protocol from biology**.

Teaching the model to anticipate the future made the vessel map matter, but timing still complicates the story.